Dr. Shanu Dengre

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Summary

Senior Data Scientist and AI Engineer (Generative AI) with 4.5+ years in production Machine Learning (ML) and a PhD in Physics. Builds Agentic AI with LangGraph LLM agents, Retrieval-Augmented Generation (RAG), prompt engineering and LLM evaluation, plus ML forecasting with 20% better accuracy across 300+ locations.

Skills

Generative AI (GenAI): LLMs, LLM APIs, Prompt/Context Engineering, RAG, LoRA, Hugging Face
Agentic AI: LangGraph, LangChain, MCP, Tool Calling, Structured Outputs, Human-in-the-loop
Retrieval & Evaluation: Embeddings, Vector Databases (FAISS, pgvector), Reranking, LLM Evaluation
ML & Deep Learning: Scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Keras, NLP, spaCy
Statistics & Forecasting: A/B Testing, Experimental Design, Time Series, TimesFM, Anomaly Detection
MLOps & Cloud: MLflow, Airflow, Docker, Kubernetes, CI/CD, Model Monitoring, AWS SageMaker, GCP
Data & Software: Python, SQL, PySpark, Snowflake, PostgreSQL, FastAPI, Kafka, Git, Linux, Agile
Computer Vision: YOLO-family, DETR, Grounding DINO, RTMDet, CNNs, CLIP, OpenCV, SAM
AI-Assisted Development: Claude (Claude Code), Gemini, GitHub Copilot
Languages: English (Fluent), German (A2.2), Hindi (Native)

Experience

InterProductTec – Mumbai *Nov 2025 – Present*
Sr. Data Scientist | **Demand Forecast Project Lead**

- Built LangGraph agents (prompt engineering, tool calling, retrieval) for NLP recipe-to-POS matching.
- Served the LLM on vLLM; agents return Pydantic structured outputs for tool calls and agent state.
- Evaluated agent outputs, adding hallucination guardrails and human-in-the-loop review of weak matches.
- Generated LLM narratives explaining forecast and KPI results to customers and operations teams.
- Led end-to-end AI demand forecasting for 300+ locations, with MLflow and Sentry model monitoring.
- Improved holiday forecast accuracy 20% with LightGBM/XGBoost, Prophet and feature engineering.
- Cut food waste 15% and lifted crew efficiency 20% across 10+ chains through demand forecasting.
- Ran model validation with backtests and A/B experiments on forecast KPIs (MAPE/WAPE, bias).
- Engineered Python, PySpark and SQL ETL pipelines on Snowflake processing millions of POS records.
- Set data science best practices and mentored junior data scientists via code and model reviews.

PreciTaste – Munich *Feb 2022 – Oct 2025*
AI Engineer and Data Scientist | **Time Series Forecasting and Computer Vision**

- Automated a DARTS MLOps pipeline with model drift detection, shortening retraining time by 50%.
- Reduced query latency 60% with FastAPI microservices on Docker/Kubernetes via GitLab CI/CD.
- Fine-tuned PyTorch YOLO CNNs and DETR/Grounding DINO transformers for food AI in 5+ chains.
- Clustered CLIP embeddings (unsupervised PCA, k-means), cutting training time 50% at equal accuracy.
- Delivered real-time POS order-matching on live Kafka with 95%+ validation accuracy, <100ms latency.

Projects

AI News Navigator – Agentic RAG, LLMs | Live Demo

- LangGraph agent (retrieve, rerank, grade, generate) with query-rewrite prompts and arXiv fallback.
- Chunked papers into a FAISS vector store for semantic search, BM25 hybrid search and bge reranking.

MCP Forecast-Drift Alert Tool | InterProductTec

- Developed an MCP-based forecasting tool that creates alerts when system accuracy drifts.

Education

PhD (Physics), TU Dresden | Publication: Phys. Rev. B 103, 064425 (2021) Oct 2015 – Feb 2021
BS-MS (Physics), IISER Pune | Publication: Phys. Rev. B 95, 054401 (2017) Aug 2010 – May 2015
Data Science & Machine Learning Bootcamp, Le Wagon Berlin June 2021 – Sept 2021